IBM BOB 2.0 HACKATHON SUBMISSION

THE ARBITER

Autonomous Forensic Compliance & Codebase Regression Arbiter

SAMEER CHAUDHARY & TEAM

DUAL-VALIDATION ENGINE

MODEL-AGNOSTIC CORE

THE INDUSTRY CRISIS

Fractured CI/CD Pipelines Breed Non-Compliance

2 Silos

Disconnected Pipeline Stages

Functional unit tests and security scans run completely isolated from each other.

Panic-Patch Violations

Urgent debugging patches frequently introduce critical PII logging leaks.

Manual Triage Exhaustion

Engineers waste hours cross-referencing git blame with compliance manuals.

CORE PARADIGM SHIFT

The Unified Forensic Philosophy

> **"Treating functional software bugs and regulatory policy violations as the exact same class of forensic problem."**

Automated evidence subpoenas, root-cause blame attribution, and a pluggable model-agnostic backend.

STATE MACHINE ARCHITECTURE

The 8-Layer Cognitive State Machine

01

Perception

Subpoenas tests, logs, secrets, and policy docs.

02

Attribution

Traces culprit commit via line-level git blame.

03

Remediation

Drafts surgical patches via FastMCP toolchain.

04

Governance

Issues auditable docket and gates approvals.

CORE INNOVATION

Adversarial Dual-Validation Engine

STANDARD AI COPILOTS

Single-Silo Validation

- Evaluates code syntax and unit tests alone.
- Blindsided by security and logging violations.
- Merges breaking compliance risks.

THE ARBITER STANDARD

Co-Equal Adjudication

- Validates functional pytest execution.
- Audits secrets and PII scanners.
- Forces auto-retry until both gates pass.

TOOL INTEGRATION

Model Context Protocol (MCP) Toolchain

Read & Write

Isolated source file patching without shell risk.

run_pytest

Executes unit suites with structured JSON output.

run_secret_scan

Detects secrets and PII patterns in source files.

git_blame

Automates line-level commit and diff tracing.

ENTERPRISE TRUST

Responsible AI Governance Framework

Human-in-the-Loop

Mandatory approval for changes touching auth or payment logic.

Confidence Thresholds

Escalates automatically if hypothesis confidence falls under 0.75.

Immutable Ledger

Append-only JSONL recording of every decision state and patch attempt.

VERIFIABLE EVIDENCE

The Tribunal Docket Deliverable

100%

Audit-Ready Record

Every incident produces a court docket linking suspect commit SHAs, scan telemetry, and parole guardrails.

Forensic Culprit Attribution

Identifies line-level commit author and diff context automatically.

Parole Conditions

Mandates regression tests and pre-commit checks before closing cases.

CORE TECHNOLOGY

Architectural Synthesis with IBM Bob 2.0

3 → 1

Autonomous Synthesis

Bob analyzed 3 competing team codebases and autonomously engineered the unified production engine.

FastMCP Protocol Integration

Zero-latency tool execution linking code analysis directly to the LLM runtime.

Model-Agnostic Provider Layer

Seamlessly connects Granite 34B, watsonx.ai, Claude, or local enterprise models.

FORENSIC DEMONSTRATION

The Multi-Concern Crime Scene

Unbounded Retry

Hangs indefinitely upon database timeout.

Hardcoded Key

AWS access keys embedded in config files.

PII Log Leak

Plaintext user email exposed in error logs.

MEASURABLE RESULTS

Enterprise Operational Impact

95%

Reduction in MTTR for combined defects.

0

Policy violations escaped to production.

<3m

Complete autonomous investigation cycle.

CASE CLOSED

VERDICT: APPROVED

Remediated, verified, and ready for human operator sign-off.

THANK YOU • QUESTIONS & LIVE DEMONSTRATION